

PropFirm Affiliate Rewards Platform

End-to-End Trader Loyalty Ecosystem, Automated S2S Verification Engine & Enterprise Admin ERP

Status: Production-Ready
Architecture: Full-Stack Web App
Security: ACID Ledger + HMAC S2S
API Docs: Swagger OpenAPI 3.0

FRONTEND

Next.js 16 +
TypeScript + Tailwind

BACKEND API

NestJS 10 +
WebSockets + JWT

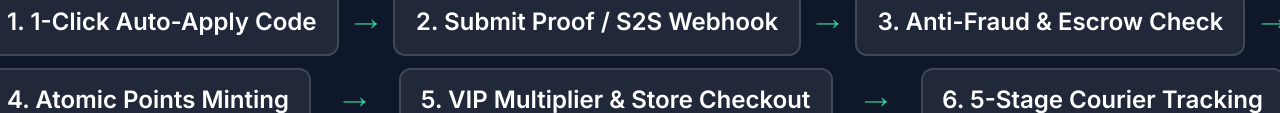
DATABASE & STORAGE

PostgreSQL (Prisma) +
Cloudflare R2

MESSAGING CHANNELS

WhatsApp Cloud API +
Resend Email

CORE TRADER-TO-REWARD LIFECYCLE



01 Public Website, Discovery & High-Conversion Tools

⚡ Cinematic Fintech UI & Social Proof

- ✓ **Dual Theme Engine:** Instant Dark & Light mode toggle with smooth transitions.
- ✓ **Trading Visuals:** Floating candlestick animations, live trading charts, and scroll progress bar.
- ✓ **Live Social Proof:** Real-time activity ticker and toast popups showing verified trader redemptions.
- ✓ **Multi-Language Support:** Built-in Google Translate switcher for global traders.

📊 Interactive Calculators & Education

- ✓ **Prop Firm & Drawdown Calculator (/compare):** Side-by-side comparison of challenge costs, discounts, drawdown rules, and points earned.
- ✓ **Interactive Evaluation Matrix:** Filter 1-Step, 2-Step, and Instant Funding tiers across partner prop firms.
- ✓ **YouTube Video Academy:** 12-video prop trading tutorial hub with built-in playback controls.
- ✓ **Full Content Suite:** Community, Trader Reviews, Announcements, FAQ, Help Center, Terms & Privacy.

- ✓ **Deep-Linked Checkout URL Builder:** Automatically injects affiliate IDs, promo codes, selected challenge tiers, and unique tracking SubIDs (PN-TRK-XXXXX) into partner checkout URLs (FTMO, Funding Pips, FundedNext, Alpha Capital, Apex, Topstep, etc.).
- ✓ **Background Clipboard Copy:** Copies the partner discount code to the user's clipboard in one click.
- ✓ **48-Hour Checkout Intent Memory:** Remembers the trader's active checkout session and automatically pre-fills the prop firm, tier, price, and promo code when they return to submit their purchase proof.

Trader Dashboard & Wallet

- ✓ **Secure Auth:** Email/Password + Google Sign-In, bcrypt hashing, and JWT token versioning.
- ✓ **Live Points Portfolio:** Tracks Available Balance, Lifetime Earned, Total Redeemed, and Pending Points.
- ✓ **Immutable Transaction History:** Full ledger audit trail of every point credit, redemption, or refund.
- ✓ **Multi-Address Book:** Save and manage multiple international shipping addresses with default selection.

3-Tier VIP Multiplier Program

VIP Tier	Points Required	Multiplier
 Rookie Trader	0 – 4,999 PTS	1.00x Base
 Funded Trader	5,000+ PTS	1.25x (+25% Boost)
 Prop Master	20,000+ PTS	1.50x (+50% + Swag)

Automated S2S Affiliate Webhooks

- ✓ **Zero-Touch Auto-Verification:** Receives real-time postback webhooks from partner affiliate networks.
- ✓ **Cryptographic Security:** Constant-time HMAC-SHA256 verification, 5-minute replay-attack window, and idempotent event keys.
- ✓ **Instant Point Minting:** Matches Order IDs, approves submissions atomically, and triggers WhatsApp + Email alerts automatically.

Anti-Fraud Proof Submission

- ✓ **Multi-File Cloud Upload:** Drag-and-drop invoice images/PDFs stored on Cloudflare R2 / S3.
- ✓ **Duplicate Claim Prevention:** Enforces unique Order ID per prop firm at the database level.
- ✓ **5-State Review Flow:** PENDING → UNDER_REVIEW → APPROVED / REJECTED / MORE_INFO_REQUIRED (with in-app trader resubmission).

Rewards Marketplace, Atomic Checkout & 5-Stage Fulfillment

- ✓ **Categorized Rewards Catalog:** Flagship Smartphones, Tablets, Trading Hardware & Monitors, Audio Gear, and Digital Gift Cards.
- ✓ **Double-Spend Protected Checkout:** Serializable ACID transaction verifies user status, validates stock, checks balance, deducts points, decrements inventory, and generates a unique redemption code (RDM-YYYY-XXXX).
- ✓ **5-Stage Courier & Voucher Tracking:** Visual delivery pipeline (PENDING → CONFIRMED → PROCESSING → SHIPPED → DELIVERED) with FedEx/DHL/UPS tracking numbers or instant digital voucher codes. Cancelled orders automatically refund points and restock inventory.

Omnichannel Communications & 24/7 Live Support Desk

Meta WhatsApp Cloud API & Email

- ✓ **Instant WhatsApp Alerts:** Automated messages for Purchase Received, Points Credited, Reward Shipped/Voucher Code, and Support Updates.
- ✓ **Outbox & Simulation Mode:** Built-in NotificationOutbox logging and simulator for pre-production testing.
- ✓ **Resend HTML Emails & In-App Bell:** Branded transactional emails and real-time in-app notification center.

24/7 Live WebSocket Chat & Ticketing

- ✓ **Real-Time Live Chat:** Floating widget and full-screen live support room powered by Socket.IO.
- ✓ **Smart Ticket Routing:** Categorized by department (PURCHASE_PROOF, CASHOUT_PAYOUT, KYC, VIP) and priority (LOW to URGENT).
- ✓ **Staff Whisper Notes:** Internal-only staff notes on tickets hidden from traders.

Enterprise Admin Control Portal & RBAC Team Management

Operations & Catalog Control

- ✓ **Executive KPI Dashboard:** Live metrics on users, pending proofs, points liability, and redemptions.
- ✓ **Verification & Fulfillment Queues:** Side-by-side receipt inspection, 1-click approve/reject, and courier tracking assignment.
- ✓ **No-Code Prop Firm & Reward CRUD:** Add/edit prop firms, challenge tiers, reward products, and stock levels dynamically.

RBAC Staff Roles & Security Audit

- ✓ **Granular Staff Roles:** SUPER_ADMIN, ADMIN, SUPPORT_LEAD, SUPPORT_AGENT, and FINANCE_OFFICER.
- ✓ **Manual Points Adjustments:** Credit or deduct user points with mandatory audit justification.
- ✓ **Immutable Audit Logs & DR:** Tracks every admin action with before/after JSON snapshots, IP addresses, plus automated balance reconciliation scripts.

